

Haar-countable sets need not form an ideal: a closed Haar-1 counterexample in $\mathbb{R}^\omega$

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Abstract

We give a candidate full negative answer to Question 7.5 of Adam Kwela’s *Haar-smallest sets* in the unrestricted class of abelian Polish groups. In the group $G = \mathbb{R}^\omega$ we construct two closed Haar-1 sets F_Y, F_Z such that $F_Y \cup F_Z$ is not Haar-countable. The construction applies a universal closed-set theorem of Banach–Głab–Jabłońska–Swaczyna to two Borel families of compacta. A Lusin–Novikov/perfect-set dichotomy then defeats every Cantor witness for the union. The result does not address the stronger possible interpretation in which the ambient group is required to be $\mathbb{R}$ or, more generally, locally compact.

1 Source and exact question

The source is Adam Kwela, *Haar-smallest sets*, arXiv:1711.09753 [1]. At the end of Section 7 the author asks:

“Question 7.5. Is the family of all Haar-countable sets an ideal?”

$(\frac{p}{q_{i_0}}, \frac{p+1}{q_{i_0}}]$, for some integer p . Thus, the distance between $\max J$ and $Z'_{i_0} + \sum_{j \leq i_0} \frac{y_j^n}{q_j}$ is at least $\frac{1}{q_{i_0}}$. Now it suffices to observe that $\sum_{j > i_0} \frac{y_j^n}{q_j} < \frac{1}{q_{i_0}}$. Hence, J is disjoint with $Z_{i_0} + \sum_{j \in \omega} \frac{y_j^n}{q_j}$, a contradiction. $\square$

We end with an open question. We believe that it requires developing new methods (besides the ones used in this paper).

Question 7.5. *Is the family of all Haar-countable sets an ideal?*

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Also, the author would like to express his gratitude to the anonymous referee for his helpful comments that improved the quality of the manuscript.

The paper works with subsets of abelian Polish groups, although many of its constructions in the surrounding sections lie in $\mathbb{R}$. We answer the unrestricted group-level formulation. Section 9 isolates the remaining scope issue.

2 Definitions and the universal tool

A *Cantor set* in a Polish space means a subspace homeomorphic to 2^ω . A subset A of an abelian Polish group X is *Haar-countable* if there are a Borel set $B \supseteq A$ and a Cantor set $C \subseteq X$ such that

$$(C + x) \cap B \text{ is countable for every } x \in X.$$

It is *Haar-1* if the same condition holds with $|(C + x) \cap B| \leq 1$ for every x . Thus every Haar-1 set is Haar-countable.

For a Polish space X , let $\mathcal{K}(X)$ be the Vietoris hyperspace of nonempty compact subsets of X . We use the following result verbatim in substance.

Theorem 1 (Universal closed-set theorem [2]). *For every nonempty analytic subspace $\mathcal{A} \subseteq \mathcal{K}(\mathbb{R}^\omega)$ there is a closed set $F \subseteq \mathbb{R}^\omega$ such that:*

- (i) *for every $K \in \mathcal{A}$ there is $a \in \mathbb{R}^\omega$ with $K + a \subseteq F$;*
- (ii) *for every $x \in \mathbb{R}^\omega$, the set $F \cap (x + [0, 1]^\omega)$ is contained in $K + d$ for some $K \in \mathcal{A}$ and $d \in \mathbb{R}^\omega$.*

This is Theorem 6.1 of [2]. Clause (i) is a universal *absorption* property; clause (ii) is a local *shape control* property. We apply the theorem twice, to complementary shapes.

3 Proof intuition

Write $G = \mathbb{R}^\omega = Y \times Z$, where Y consists of the even coordinates and Z of the odd coordinates. One universal closed set will locally resemble compact graphs over Z ; a Cantor set moving only in the Y coordinates therefore meets each translate in at most one point. The other universal set interchanges Y and Z .

Why can their union be large? Every Cantor set $C \subseteq Y \times Z$ contains a Cantor subset which is a graph in one of the two coordinate directions. If an uncountable Z -fiber exists, use a Cantor subset of that fiber, on which the Y -projection is injective. If all Z -fibers are countable, Lusin–Novikov decomposes C into countably many graphs over Z ; one graph is uncountable and contains a Cantor set. Universal absorption then translates this Cantor subset into one of the two closed sets. Thus every candidate witness has a translate with uncountable intersection with the union.

4 The two analytic hyperspace families

Let $\pi_Y : G \rightarrow Y$ and $\pi_Z : G \rightarrow Z$ be the coordinate projections, and put

$$\mathcal{K}_Z = \{K \in \mathcal{K}(G) : \pi_Z|_K \text{ is injective}\}, \quad \mathcal{K}_Y = \{K \in \mathcal{K}(G) : \pi_Y|_K \text{ is injective}\}.$$

Lemma 1. *The families $\mathcal{K}_Y$ and $\mathcal{K}_Z$ are nonempty Borel subspaces of $\mathcal{K}(G)$.*

Proof. Nonemptiness is immediate, for instance from singletons. We prove the Borel claim for $\mathcal{K}_Z$; the other case is symmetric. Fix compatible metrics d on G and d_Z on Z . For $r, s \geq 1$, let

$$U_{r,s} = \left\{ K \in \mathcal{K}(G) : \forall u, v \in K, \quad d(u, v) < \frac{1}{r} \text{ or } d_Z(\pi_Z u, \pi_Z v) > \frac{1}{s} \right\}.$$

For fixed r, s , the displayed disjunction defines an open subset of $G \times G$. The set of compacta K whose square is contained in that open set is open in the Vietoris hyperspace: compactness of K^2 gives a finite product-neighborhood refinement which persists for all sufficiently close compacta. Hence every $U_{r,s}$ is open.

We claim

$$\mathcal{K}_Z = \bigcap_{r \geq 1} \bigcup_{s \geq 1} U_{r,s}.$$

If $\pi_Z|_K$ is injective, then on the compact set $E_r = \{(u, v) \in K^2 : d(u, v) \geq 1/r\}$ the continuous function $d_Z(\pi_Z u, \pi_Z v)$ is strictly positive and therefore has a positive minimum. A suitable s gives $K \in U_{r,s}$. Conversely, if two distinct points $u, v \in K$ have the same Z -projection, choose r with $d(u, v) \geq 1/r$; then $K \notin U_{r,s}$ for every s . The claimed formula follows, so $\mathcal{K}_Z$ is Borel and in particular analytic. $\square$

Apply Theorem 1 to $\mathcal{K}_Z$ and $\mathcal{K}_Y$. We obtain closed sets $F_Z, F_Y \subseteq G$ such that, for $i \in \{Y, Z\}$,

$$\text{every } K \in \mathcal{K}_i \text{ has a translate contained in } F_i, \tag{1}$$

$$F_i \cap (x + [0, 1]^\omega) \subseteq K + d \text{ for some } K \in \mathcal{K}_i, d \in G. \tag{2}$$

5 Each universal set is Haar-1

Identify Y and Z with their corresponding coordinate subgroups of G . Inside $Q = [0, 1]^\omega$, define

$$C_Y = \{x \in G : x(2n) \in \{0, 1\}, x(2n+1) = 0 \text{ for all } n\},$$

and define C_Z symmetrically, with the odd coordinates varying in $\{0, 1\}$ and the even coordinates equal to zero. Both are Cantor sets.

Lemma 2. *F_Z is Haar-1, witnessed by C_Y , and F_Y is Haar-1, witnessed by C_Z .*

Proof. Fix $x \in G$. Since $C_Y + x \subseteq Q + x$, the localization property (2) gives

$$(C_Y + x) \cap F_Z \subseteq (C_Y + x) \cap (K + d)$$

for some $K \in \mathcal{K}_Z$ and $d \in G$. Every point of $C_Y + x$ has the same Z -projection, while π_Z is injective on $K + d$. The intersection on the right therefore has at most one point. Since F_Z is closed and x was arbitrary, C_Y witnesses that F_Z is Haar-1. Interchanging Y and Z proves the assertion for F_Y . $\square$

6 Every Cantor set contains one of the two shapes

Lemma 3 (Coordinate dichotomy). *Every Cantor set $C \subseteq Y \times Z$ contains a Cantor set $K \in \mathcal{K}_Y \cup \mathcal{K}_Z$.*

Proof. Consider the fibers of $\pi_Z|_C : C \rightarrow Z$.

First suppose that some fiber

$$C_z = C \cap (Y \times \{z\})$$

is uncountable. It is a closed subset of the compact metrizable space C , hence an uncountable Polish space. By the perfect set theorem it contains a Cantor set K . The map π_Y is injective on C_z , so $K \in \mathcal{K}_Y$.

Now suppose every fiber of $\pi_Z|_C$ is countable. Regard C , after swapping the coordinates, as a closed relation in $Z \times Y$ with countable vertical sections. The Lusin–Novikov uniformization theorem decomposes C into countably many Borel partial graphs. Equivalently, there are Borel sets $B_n \subseteq C$ with

$$C = \bigcup_{n \in \mathbb{N}} B_n$$

such that π_Z is injective on each B_n . Since C is uncountable, some B_n is uncountable. An uncountable Borel subset of a Polish space contains a Cantor set K . Then $\pi_Z|_K$ is injective, so $K \in \mathcal{K}_Z$. $\square$

7 The counterexample

Theorem 2. *In the abelian Polish group $G = \mathbb{R}^\omega$ there are two closed Haar-1 sets whose union is not Haar-countable. Consequently, Haar-countable subsets do not form an ideal in the unrestricted class of abelian Polish groups.*

Proof. Take the closed Haar-1 sets F_Y, F_Z constructed above. We show that $F = F_Y \cup F_Z$ is not Haar-countable.

Let $C \subseteq G$ be an arbitrary Cantor set. By Lemma 3, C contains a Cantor set $K \in \mathcal{K}_Y \cup \mathcal{K}_Z$. If $K \in \mathcal{K}_i$, the absorption property (1) gives a translation vector $a \in G$ such that $K + a \subseteq F_i$. Therefore

$$(C + a) \cap F \supseteq K + a,$$

and the intersection is uncountable.

This argument applies to every Cantor set C . Moreover, if B is any Borel hull of F , then $(C + a) \cap B$ contains the same uncountable set $K + a$. Thus no Cantor set can witness Haar-countability of F , and $F \notin \text{HCtbl}$. On the other hand, Lemma 2 gives $F_Y, F_Z \in \text{HCtbl}$. Hence HCtbl is not closed under finite unions and is not an ideal. $\square$

The conclusion is stronger than the minimum required for non-ideality: both summands are closed and Haar-1, the smallest nontrivial level in the hierarchy.

8 Compatibility with the source's Theorem 2.3

The source paper prints a statement, Theorem 2.3, asserting that a Haar- n set and a Haar- m set have Haar- $(n + m)$ union. Read literally for arbitrary subsets of arbitrary abelian Polish groups, that statement would contradict Theorem 2. The proof printed immediately below Theorem 2.3, however, invokes the fact that σ -compact Haar- k sets are generically Haar- k and then applies this genericity conclusion to the two sets without having assumed σ -compactness.

The cited background result appears in the present version of [2] as Theorem 16.5(3), and it explicitly has the σ -compact hypothesis. Closed subsets of the non-locally-compact Polish group $\mathbb{R}^\omega$ need not be σ -compact, and the universal construction supplies no such conclusion for F_Y or F_Z . Thus the counterexample is consistent with the proved genericity theorem and exposes the missing hypothesis in the broader printed finite-union assertion. This point deserves particular human scrutiny because it is also an independent correction to the source paper.

9 Scope, novelty audit, and review status

Scope. The theorem gives a complete negative answer if Question 7.5 asks whether Haar-countable sets form an ideal in every abelian Polish group. It does not construct a counterexample in $\mathbb{R}$. The proof uses the non-locally-compact group $\mathbb{R}^\omega$ and a universal theorem special to that setting. No relevant topological group embedding into $\mathbb{R}$ exists, and arbitrary Borel coding does not preserve Haar-countability. Therefore the real-line or locally compact version, if that was the intended question, remains open in this packet.

Version and literature audit. Bounded searches on 13–17 August 2026 used the exact question, the source title and arXiv identifier, the author's name, and combinations of *Haar-countable*, *ideal*, and $\mathbb{R}^\omega$. No later paper or note explicitly answering Question 7.5 was found. The current arXiv abstract says that Haar-countable sets do not form an ideal, while the current source TeX, the PDF, and the published paper still pose Question 7.5. We treat the abstract sentence as a metadata inconsistency rather than as a documented solution. The universal theorem itself is published and cited below; the application through the two coordinate-injective hyperspace families was not found in the bounded search.

Status and human review. This packet is a candidate full, likely valid counterexample for the unrestricted abelian-Polish-group formulation. The four points to check most carefully are Lemma 1, the two applications of Theorem 1, the Lusin–Novikov step in Lemma 3, and the intended ambient-group scope of Question 7.5. Section 8 should also be checked against the precise version history of the source's cited background theorem.

References

- [1] Adam Kwela, *Haar-smallest sets*, arXiv:1711.09753; Topology and its Applications 267 (2019), 106895.
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- [3] Alexander S. Kechris, *Classical Descriptive Set Theory*, Graduate Texts in Mathematics 156, Springer, 1995.